Notas del Curso de Métodos Numéricos

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Capítulo 1

Factorización LU

1.1. Ejemplo 1

```r
A=matrix(c(1,1,1,1,
           2,3,1,5,
           -1,1,-5,3,
           3,1,7,-2),byrow = TRUE,nrow = 4)
(A)
```

```
##      [,1] [,2] [,3] [,4]
## [1,]    1    1    1    1
## [2,]    2    3    1    5
## [3,]   -1    1   -5    3
## [4,]    3    1    7   -2
```

```r
b=matrix(c(10,31,-2,18),nrow = 4)
(b)
```

```
##      [,1]
## [1,]   10
## [2,]   31
## [3,]   -2
## [4,]   18
```

```r
Ab <- cbind(A,b); (Ab)
```

```
##      [,1] [,2] [,3] [,4] [,5]
## [1,]    1    1    1    1   10
## [2,]    2    3    1    5   31
## [3,]   -1    1   -5    3   -2
## [4,]    3    1    7   -2   18
```

Los pivotes se definen como $a_{kk}^{(k)}$, luego construimos los multiplicadores: $l_{i,k} = a_{ik}^{(k)}/a_{kk}^{(k)}$, $l_{21} = a_{21}/a_{aa}$ y $l_{31} = a_{31}/a_{11}$ y $l_{41} = a_{41}/a_{11}$

```r
A <- Ab;
l21 <- A[2,1]/A[1,1];(l21)
```

```
## [1] 2
```

```r
l31 <- A[3,1]/A[1,1];(l31)
```

```
## [1] -1
```

```r
l41 <- A[4,1]/A[1,1];(l41)
```

```
## [1] 3
```

Construimos la matriz U, donde las entradas $a_{ij}^{(k+1)} = a_{ik}^{(k)} - l_{ik} \times a_{kj}^{(k)}$

```r
A[2,] <- A[2,]-l21*A[1,]; (A[2,])
```

```
## [1]  0  1 -1  3 11
```

```r
A[3,] <- A[3,]-l31*A[1,]; (A[3,])
```

```
## [1]  0  2 -4  4  8
```

```r
A[4,] <- A[4,]-l41*A[1,]; (A[4,])
```

```
## [1]   0  -2   4  -5 -12
```

Es decir la matriz resultante es:

```r
(A)
```

```
##      [,1] [,2] [,3] [,4] [,5]
## [1,]    1    1    1    1   10
## [2,]    0    1   -1    3   11
## [3,]    0    2   -4    4    8
## [4,]    0   -2    4   -5  -12
```

entonces el pivote es $A_{22} = 1$, calculemos los l_{32} y l_{42}

```r
l32 <- A[3,2]/A[2,2];(l32)
```

```
## [1] 2
```

```r
l42 <- A[4,2]/A[2,2];(l42)
```

```
## [1] -2
```

Haciendo cero debajo del pivote

```r
A[3,] <- A[3,]-l32*A[2,];(A[3,])
```

```
## [1]   0   0  -2  -2 -14
```

```r
A[4,] <- A[4,]-l42*A[2,];(A[4,])
```

```
## [1]  0  0  2  1 10
```

La matriz A queda de la forma:

```r
(A)
```

```
##      [,1] [,2] [,3] [,4] [,5]
## [1,]    1    1    1    1   10
## [2,]    0    1   -1    3   11
## [3,]    0    0   -2   -2  -14
## [4,]    0    0    2    1   10
```

por tanto el pivote es $A_{33} = -2$, y resta calcylar l_{43}

```
l43 <- A[4,3]/A[3,3];(l43)
```

```
## [1] -1
```

haciendo cero debajo del pivote

```
A[4,] <- A[4,]-l43*A[3,];(A[4,])
```

```
## [1]  0  0  0 -1 -4
```

Por lo tanto la matriz A resultante es

```
(A)
```

```
##      [,1] [,2] [,3] [,4] [,5]
## [1,]    1    1    1    1   10
## [2,]    0    1   -1    3   11
## [3,]    0    0   -2   -2  -14
## [4,]    0    0    0   -1   -4
```

entonces podemos construir la matriz L con los valores l_{ij} calculados en los pasos anteriores

```
L=diag(4);(L)
```

```
##      [,1] [,2] [,3] [,4]
## [1,]    1    0    0    0
## [2,]    0    1    0    0
## [3,]    0    0    1    0
## [4,]    0    0    0    1
```

```
L[2,1] <- l21
L[3,1] <- l31
L[4,1] <- l41
L[3,2] <- l32
L[4,2] <- l42
L[4,3] <- l43
(L)
```

```
##      [,1] [,2] [,3] [,4]
## [1,]    1    0    0    0
## [2,]    2    1    0    0
## [3,]   -1    2    1    0
## [4,]    3   -2   -1    1
```

```
U <- A[,1:4];(U)
```

```
##      [,1] [,2] [,3] [,4]
## [1,]    1    1    1    1
## [2,]    0    1   -1    3
## [3,]    0    0   -2   -2
## [4,]    0    0    0   -1
```

Verifiquemos que efectivamente $A = LU$

```
Acalculada <- L%*%U; (Acalculada)
```

```
##      [,1] [,2] [,3] [,4]
## [1,]    1    1    1    1
```

```
## [2,]    2    3    1    5
## [3,]   -1    1   -5    3
## [4,]    3    1    7   -2
```

1.2. Ejemplo 2

Apliquemos lo desarrollado para la siguiente matriz A

```
A=matrix(c(4,0,1,1,
           3,1,3,1,
           0,1,2,0,
           3,2,4,1),nrow = 4,byrow = TRUE);
(A)
```

```
##      [,1] [,2] [,3] [,4]
## [1,]    4    0    1    1
## [2,]    3    1    3    1
## [3,]    0    1    2    0
## [4,]    3    2    4    1
```

```
b=matrix(c(5,6,13,1),nrow = 4);(b)
```

```
##      [,1]
## [1,]    5
## [2,]    6
## [3,]   13
## [4,]    1
```

1)

```
Ab <- cbind(A,b)
A <- Ab;
```

2)

```
l21 <- A[2,1]/A[1,1];(l21)
```

```
## [1] 0.75
```

```
l31 <- A[3,1]/A[1,1];(l31)
```

```
## [1] 0
```

```
l41 <- A[4,1]/A[1,1];(l41)
```

```
## [1] 0.75
```

3)

```
A[2,] <- A[2,]-l21*A[1,]; (A[2,])
```

```
## [1] 0.00 1.00 2.25 0.25 2.25
```

```
A[3,] <- A[3,]-l31*A[1,]; (A[3,])
```

```
## [1]  0  1  2  0 13
```

```
A[4,] <- A[4,]-l41*A[1,]; (A[4,])
```

```
## [1]  0.00  2.00  3.25  0.25 -2.75
```

4)

```
l32 <- A[3,2]/A[2,2];(l32)
```

```
## [1] 1
```

```
l42 <- A[4,2]/A[2,2];(l42)
```

```
## [1] 2
```

5)

```
A[3,] <- A[3,]-l32*A[2,];(A[3,])
```

```
## [1]  0.00  0.00 -0.25 -0.25 10.75
```

```
A[4,] <- A[4,]-l42*A[2,];(A[4,])
```

```
## [1]  0.00  0.00 -1.25 -0.25 -7.25
```

6)

```
l43 <- A[4,3]/A[3,3];(l43)
```

```
## [1] 5
```

7)

```
A[4,] <- A[4,]-l43*A[3,];(A[4,])
```

```
## [1]   0   0   0   1 -61
```

8)

```
L=diag(4);(L)
```

```
##      [,1] [,2] [,3] [,4]
## [1,]    1    0    0    0
## [2,]    0    1    0    0
## [3,]    0    0    1    0
## [4,]    0    0    0    1
```

9)

```
L[2,1] <- l21
L[3,1] <- l31
L[4,1] <- l41
L[3,2] <- l32
L[4,2] <- l42
L[4,3] <- l43;
(L)
```

```
##      [,1] [,2] [,3] [,4]
## [1,] 1.00    0    0    0
## [2,] 0.75    1    0    0
## [3,] 0.00    1    1    0
## [4,] 0.75    2    5    1
```

10)

```
U <- A[,1:4];(U)
```

```
##      [,1] [,2]  [,3]  [,4]
## [1,]    4    0  1.00  1.00
## [2,]    0    1  2.25  0.25
## [3,]    0    0 -0.25 -0.25
## [4,]    0    0  0.00  1.00
```

11)

```
Acalculada <- L%*%U; (Acalculada)
```

```
##      [,1] [,2] [,3] [,4]
## [1,]    4    0    1    1
## [2,]    3    1    3    1
## [3,]    0    1    2    0
## [4,]    3    2    4    1
```